Oct 19 — Day summary

IAA AI-Assisted Hackathon 2026 · Woojin Ji & Krista Stanke · Part 1 closes Oct 23

> **Bottom line.** First Kaggle submissions made. The adopted model moved from 0.11803 to **0.11907** while the target definition was corrected; an experimental blend reached 0.12186 (not adopted). Validation predicted the leaderboard change four times in a row, so decisions stay on validation.

Public leaderboard (top 10 on Oct 19: 0.125–0.130)

File	What changed	Score
glm_v2	Ridge GLM per line (drop=None), balance correction; no excess on liability	0.11803
glm_v2_excess_liability	Excess deducted on every peril (correct definition)	0.11724
glm_v3	Unstable features removed (previous_insurer; occupation, region in home lines); duplicate customer_tenure removed	0.11798
glm_v4 (adopted)	**v3 + claim history (same policy, claims in the previous 1 and 2 years)**	**0.11907**
blend_v1 / v2 (experiments)	GLM + LightGBM 50/50 in buildings and contents; fail the tail condition	0.12113 / 0.12186

What we did

- **Regularisation.** Ridge with every level coded removed the lowest-decile under-pricing and the buildings 2024 over-prediction. Chosen: motor α0.5/10 bands, buildings α0.5/20, contents α0.3/20. Validation D^2: 0.050→0.051, 0.055→0.065, 0.060→0.067.
- **Balance correction.** A log-link Tweedie GLM does not reproduce the training total, and ridge widens the gap. One factor k per model, estimated on training years only, restores the level (buildings validation A/E 0.99–1.02).
- **Excess definition.** The host confirmed the excess applies to every peril. Default changed in `src/target.py`, five test expectations recalculated by hand, downstream steps re-checked.
- **Feature audit.** Rank correlation of one-way cost between 2021–22 and 2023–24: previous_insurer ≈ 0 in all lines → removed; GLM improved. Associations: renewal_count = customer_tenure (1.00), driver_age ≈ years_licensed (0.99), region ≈ weather (η 0.83–0.94).
- **LightGBM challenger.** In home lines GBM/blend gain +0.006–0.008 D^2 on validation, but over-price the lowest decile (A/E 0.64–0.81). Not adopted.
- **Claim history.** Rows with a claim in the previous year were under-priced by 22–80%. Added as a feature (`src/history.py`, 5 tests): motor A/E for those rows 1.22→1.05/0.98. Self-reported prior_claims_5y is unrelated to recorded claims (Spearman 0.04).

Checks on AI output today

AI output	Outcome
Excess does not apply to liability perils (UK practice)	wrong host answer; the leaderboard had favoured the wrong reading
Frequency re-check counting only net > 0 claims	false alarm three definitions side by side; 0.93 shift confirmed
Tie-break rule; 3% limit on k; level check against all-year average	corrected rule direction, out-of-sample check, latest-year comparison
Off-balance factor; claim-history signal; leaderboard predictions	confirmed by validation and four leaderboard checks

Open issues

- Motor 2023 validation A/E still 1.029; glass_hail frequency rises ≈3% a year (not adjusted); recoveries on open claims.
- Policy limits: claims vs sum insured not yet checked (notebook 12).
- discount_applied (linked to channel) — possible pricing leakage; age_at_licence as a feature.

Next

- **Tonight:** results of the unattended run (NCB check, method B = GBM on top of GLM v4, GBM settings chosen on the last training year); limits check.
- **Tue–Wed:** large-loss loading by risk, richer history (peril, amount), motor BI split if time; feature_audit.py for Week 2.
- **Thu:** final model fixed, pipeline script, report skeleton. **Fri Oct 23:** submit and select final entries by validation, not by public score.